

Mastering PandaBoard



Part—1

Over the next couple of months, this series of articles aims to make a novice well-versed with PandaBoard. We will cover various topics such as porting different OSs to PandaBoard, image processing with it, and so on. This first article gives a brief overview of PandaBoard and how to install Xubuntu on it.

PandaBoard (Figure 1) is a low-cost, open, OMAP 4-based mobile software development platform that can serve as a fast ARM-based single-board computer. In short, you can do some awesome stuff with it. Amazed? Yes, this small board, weighing less than 100g, is packed with powerful features such as:

- The OMAP4430 application processor consisting of a dual-core 1 GHz ARM Cortex-A9 MPCore CPU, 304 MHz PowerVR SGX540 GPU and a C64x DSP
- 1 GB low-power DDR2
- Display HDMI v1.3 connector (Type A) to drive HD displays and a DVI-D connector
- 3.5 mm audio in/out and HDMI audio out
- Full-size SD/MMC card
- Built-in 802.11 and Bluetooth v2.1+EDR
- On-board 10/100 Ethernet
- Expansion: 1xUSB OTG, 2xUSB HS host ports and general-purpose expansion header
- Size: 100 × 110 mm (4 × 4.5 inches)

PandaBoard is such a powerful mobile computing platform that one can port various OSs to it. Already supported are Ubuntu, Xubuntu, Android, Angstrom, Meego, Fedora, etc.

Where to buy it

There are a lot of distributors, but I recommend buying from SVTronics (<http://www.svtronics.com/ecommerce/panda>), the manufacturers of PandaBoard. The current cost is \$174.

Accessories

I would also suggest that interested readers buy the following accessories, which I used:

- A HDMI/DVI cable to see the output on a monitor

- A 8/16 GB SDHC or micro SD card (then use it with a converter)
- A 5V power supply
- A USB hub
- A serial-to-USB cable
- The keyboard
- The mouse
- A laptop cooler (so that PandaBoard does not heat up)

Install image

In this article, we will try to install Ubuntu 12.04 LTS; please visit <http://cdimage.ubuntu.com/releases/12.04/release/> and download the Texas Instruments OMAP4 (Hard-Float) pre-installed desktop image.

Prepare the SD card

I'm using Ubuntu 11.10 on a 64-bit (desktop) machine; I suggest you use the Ubuntu environment too, if possible. If you are not familiar with Ubuntu, don't worry—watch my video 'Getting Started With Ubuntu' at <http://www.youtube.com/watch?v=6Cb-227Ogzw>.

Now insert your SD card into the desktop, but make sure that it is not mounted. Identify the correct raw device name—for the card, not the partition (like `/dev/sdb` and not `/dev/sdb1`). You could use the Ubuntu disk utility and select the SD card. In a terminal, navigate to the folder containing the downloaded Ubuntu image, and to write it to the SD card, run the following commands:

```
zcat ./ubuntu-12.04-preinstalled-desktop-armhf+omap4.img.gz |sudo
dd bs=4M of=/dev/sdb
sudo sync
```